Theory 13.4.1

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1 Equivariance and invariance

If $\mathbf{P}$ is a permutation matrix, then the following holds:

$$\mathbf{H}_{k+1}\mathbf{P} = \mathbf{F}[\mathbf{H}_k\mathbf{P}, \mathbf{P}^T\mathbf{A}\mathbf{P}, \boldsymbol{\phi}_k].$$

2 Problem 13.7

Consider the simple GNN layer:

$$\mathbf{H}_{k+1} = GraphLayer[\mathbf{H}_k, \mathbf{A}] = \mathbf{a}[\boldsymbol{\beta}_k\mathbf{1}^T + \boldsymbol{\Omega}_k \begin{bmatrix} \mathbf{H}_k \\ \mathbf{H}_k\mathbf{A} \end{bmatrix}]$$

where H is a $D \times N$ matrix containing the N node embeddings in its columns, A is the $N \times N$ adjacency matrix, β is the bias vector, and Ω is the weight matrix.

Show that

$$GraphLayer[\mathbf{H}_k, \mathbf{A}]\mathbf{P} = GraphLayer[\mathbf{H}_k\mathbf{P}, \mathbf{P}^T\mathbf{A}\mathbf{P}]$$